FLEET TEAM DIGEST

737MAX-FTD-21-21002

Issue Title : Part Number (P/N) 63396754-2 Electronic Flow Control (Eflow) System - Unexpected Air Conditioning Flow Control Valve Closures in Flight

Airplane Model	ATA	Minor Model(s)
737MAX	2150-00	-8, -8200, -9
Other Model(s)	**ECCN**	
737NG	9E991	

Originated Date	Last Revised Date	Created On
07/29/2021	10/25/2022	10/14/2022
Estimated Completion Date	**Next Update Date**	

Revision Description

Re-opened this FTD article to highlight the following:

- Deleting reference to 737-SL-21-117 service letter as it is not applicable to MAX aircraft equipped with IASCs, which unlike the NG, include logic that can detect eFCV faults for various valve closure conditions.
- MAX AHM alerts 21-0011-C730 / 21-0012-C730 have been deleted to prevent unnecessary -2 eFCV removals not specifically associated with MMSGs 21-55030/55031 triggered.

This FTD article is considered closed based on information reflected in the Final Action section.

All changes are higlighted in blue text.

Applicability

All 737MAX airplanes

Description

As noted in Fleet Team Digest (FTD) articles 737NG-FTD-21-16006 and 737NG-FTD-21-18001, electronic flow control valve (eFCV) Part Number (P/N) 63396754-2 was introduced into 737NG production on aircraft L/N 6361 in April 2017. Subsequently, all 737MAX airplanes were delivered with the -2 eFCV.

The -2 eFCV incorporated a new servo and torque motor, which mitigated the “unexpected eFCV closure” phenomena as a result of P/N 63396754-1 valve’s susceptibility to a stuck reference regulator. Unlike the -1 eFCV whose stuck reference regulator results in unexpected closure of the valve during some bleed pressure conditions, the -2 eFCV was designed to fail in the FULL OPEN position when the reference regulator gets stuck.

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Upon introduction into the fleet, 737NG operators began to report indications of unexpected -2 eFCV closures in flight from Airplane Condition Monitoring System (ACMS) alerts. Similar to the -1 eFCV, temporary -2 eFCV closures were also observed during bleed pressure transients (example: following a rise in bleed duct pressure such as during takeoff; or other times during a flight when the throttle is adjusted rapidly up or down). The majority of the -2 eFCV closure indication events were also reported by operators in the Asia-Pacific region at the onset of this issue.

Boeing notes that the -2 eFCV failure mode differs from that exhibited on the -1 valve, as described in FTD article 737NG-FTD-21-16006. None of the -2 eFCV closures have resulted in a cabin depressurization event, as there have been no reports of aircraft losing both packs in flight without indication. Momentary -2 eFCV closures have only been shown to occur on one pack side, thereby leaving the operative pack fully capable of maintaining pressurization.

737NG AHM Alerts 21-0011-S611 / 21-0012-S611 were established by Boeing to monitor unexpected eFCV closures in flight in accordance with service letter 737-SL-21-117 guidance. Equivalent MAX AHM Alerts 21-0011-C730 / 21-0012-C730 were carried over during the initial -2 eFCV closure investigation.

NOTE: Unlike the 737NG model, the Integrated Air System Controller (IASC) on 737 MAX aircraft contains logic that can detect various FCV faults (i.e. FCV FAIL OPEN, FCV FAIL CLOSED, FCV STUCK CLOSED).

Background

Boeing does not consider temporary -2 eFCV closures that occur during bleed pressure transients to be anomalous. Please note that both -1 and -2 eFCVs may close during a bleed pressure transient, but a failed -1 eFCV does not recover until the bleed pressure is decreased much lower than the pressures observed with the -2 eFCV temporary closures. All reported -2 eFCV closures have recovered when bleed pressures were still high. The event data related to uncommanded temporary closures of the -2 eFCV indicates that, for closures due to bleed transients, the closure duration is approximately 6 seconds on average, but consistently less than 12 seconds.

At the onset of the Honeywell/Boeing joint investigation, the team performed multiple system-level tests at Honeywell's Torrance, California facility in hopes of duplicating these reported valve closure events. Numerous teardown investigations have been performed but none have revealed the root cause for the unexpected valve closure phenomena. While other anomalies have been identified during failed incoming Acceptance Test Procedure (ATP) tests, none of these discrepancies can be correlated to the uncommanded valve closure issue. Note, these events were never duplicated by the Honeywell 737NG/MAX eFlow flow model.

In June 2019, Boeing and Honeywell met at the supplier's Tempe (Arizona) test facility to perform a detailed teardown investigation (to also include specific As-Received testing, hysteresis analyses, and CT scanning prior to valve disassembly) on two removed, serviceable, and “previously-modified” P/N 63396754-2 eFCVs, which had exhibited the un-commanded closure phenomenon. The technician pressure tested the eFCV's piston, while the valves were still fully assembled. No anomalies were detected on either valve.

Status

Boeing and Honeywell have concluded their root cause investigation into the -2 eFCV closure phenoma for the following reasons:

- Minimal reporting activity of such events from the fleet since 2019

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- Note: The Integrated Air System Controller (IASC) on 737MAX aircraft contains logic that can detect various FCV faults (i.e. FCV FAIL OPEN, FCV FAIL CLOSED, FCV STUCK CLOSED). No significant issues occurring in the fleet that create a maintenance burden or impact dispatch operations.

- None of the -2 eFCV closures have resulted in a cabin depressurization event (i.e. data substantiates that the remaining opertive pack is capable of maintaining pressurization).
- Unable to replicate eFCV-related faults and identify root cause for a) this unexpected closure condition or b) the '737MAX PACK Light Illumination After Engine Start' issue discussed in 737MAX-FTD-21-18001

Since the MAX IASC logic can detect eFCV faults for various valve closure conditions inflight, the reporting of MAX AHM alerts 21-0011-C730 / 21-0012-C730 (Unexpected RH / LH air conditioning pack EFCV closure inflight) without annunciation of a Pack Flow Control Valve status message (with correlated MMSGs 21-55030 / 21-55031) will result in unnecessary -2 eFCV removals. As a result, MAX AHM alerts 21-0011-C730 / 21-0012-C730 have been deleted to prevent unnecessary valve removals not specifically tied to MMSGs 21-55030/55031. In addition, Boeing is deleting service letter 737-SL-21-117 as a referenced document within this FTD article since it is applicable to NG aircraft only.

With regard to 737MAX MMEL/DDG dispatch criteria, Boeing has made the following changes:

- Added MMEL requirement for Sub-item 21-51-02-01 OPERATIONS (O) “Electronic Flow Control” procedure with a condition to require the application of performance adjustments when the backup pneumatic flow control mode takes over to compensate for failure of electronic flow control (via failure of the eFCV torque motor or primary PFTC) of one or both air conditioning packs. Note, the backup pneumatic flow mode has a higher flow set point than the torque motor controlled flow inherent to a properly functioning eFCV. Status message PACK FLOW PRIMARY L or PACK FLOW PRIMARY R will be present to confirm this condition exists.
- In accordance with the information published in FTD article 737MAX-FTD-21-21005, the FAA approved Boeing's proposed revisions to 737MAX MMEL/DDG 21-51-02-01 to restrict cargo carriage in the aft compartment as mandated by FAA AD 2021-16-15. The MMEL was issued in December 2021, and the associated DDG was issued in January 2022. Operators can track the progress of all activities associated with FTD article 737MAX-FTD-21-21005 in future updates.
- Created Item No. 21-51-02-02 “Pack Flow Control and Shutoff Valves Position Indicator with Discrete Signal” with options A (Discrete Signal Failed Open) and B (Discrete Signal Failed Closed).

Interim Action

None.

Final Action

- With little to no reports from the fleet over recent years, Boeing and Honeywell have opted to close out this issue with no planned corrective action(s) at this time.
- Continue to follow IFIM tasks 21-51-00-810-807 / 808 in accordance with observed Maintenance Messages 21-55030 (Left A/C Pack FCV Signal is Out of Range or Valve is not in Commanded Position) and 21-55031 (Right A/C Pack FCV Signal is Out of Range or Valve is not in Commanded Position), respectively.

  - If the condition described in 737MAX-FTD-21-18001 “737MAX PACK Light Illumination After Engine Start” occurs where the eFCV does not open when commanded (showing MMSG 21-55030 or 21-55031), please report to Boeing.

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- 737MAX MMEL/DDG 21-51-02-01 dispatch criteria revisions (i.e. accounting for backup pneumatic flow mode) completed in December 2021 / January 2022

Milestones

Initial Reports of -2 eFCV Closures: June 2017

Boeing / Honeywell Root Cause Investigation Initiated: June 2017

Boeing / Honeywell Teardown Investigations (Tempe, AZ test facility): June 2019 (complete)

737MAX MMEL and DDG 21-51-02-01 / -02 Updates: 30-April-2021 (complete)

737MAX MMEL/DDG 21-51-02-01 Dispatch Criteria Updates (i.e. Accounts for backup pneumatic flow mode): December 2021 / January 2022 (complete)
Deletion of MAX AHM alerts 21-0011-C730 / 21-0012-C730: October 2022 (complete)

Operator Action

Refer to the Final Action section.

Supplier Information

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Website: www.myaerospace.com

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737MAX-FTD-21-21005		

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Fleet Team Digest Article (FTD)	737NG-FTD-21-16006		
Component Maintenance Manual (CMM)	21-10-45	4	03/31/2017
Minimum Equipment List (MEL)	21-51-02-01		
Minimum Equipment List (MEL)	21-51-02-02		
Fleet Team Digest Article (FTD)	737NG-FTD-21-18001		
Fleet Team Digest Article (FTD)	737MAX-FTD-21-18001		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Flow Control Valve		Honeywell	63396754-2

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- CRC - Component Reliability Collaboration : Collaborative process to eliminate component removals in production and the during the first year of new airplane in service

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